units, and support infrastructure, limiting adoption by smaller organizations or research institutions.

- *Limited multi-mission flexibility*: Many UAVs were designed for specific mission profiles, such as reconnaissance or target simulation. Payloads were not easily interchangeable, reducing adaptability for diverse applications.
- *Vulnerability to electronic countermeasures*: With increasing radar and electronic warfare capabilities, second-generation UAVs were susceptible to jamming, signal spoofing, or interception, compromising mission integrity and data security.

3.7 Case Study: Operational Deployment and Impact of Second-generation Drones

The following case studies illustrate the technological evolution, strategic value, and real-world deployment of second-generation drones through their use in key military operations and reconnaissance missions. These case studies highlight how UAVs transitioned from experimental platforms to indispensable tactical *assets* capable of transforming battlefield awareness, mission precision, and data-driven decision-making.

3.7.1 Case study 1: Ryan Model 147 “Lightning Bug” – reconnaissance in the Vietnam War (1964–1975)

The Ryan Model 147 “Lightning Bug,” a derivative of the Ryan Firebee, was among the first UAVs to demonstrate large-scale operational success during the Vietnam War. Deployed by the US Air Force under the *Buffalo Hunter Program*, these drones conducted more than 3000 reconnaissance missions, collecting photographic intelligence deep within enemy territories, including Vietnam, Laos, China, and North Korea.